

Arnav Kakkar

+91 9996438080 arnavkakk28.3@gmail.com
in arnavkakk arnavkakk

Education

Thapar Institute of Engineering and Technology, Patiala — B.E. Electronics Engineering Sep 2023 – Jun 2027
The Vivekanand School, New Delhi — CBSE Senior Secondary Apr 2021 – Jun 2023

Experience

Network Data Analytics Intern Jun 2026 – Jul 2026

2WAY Technologies Pvt. Ltd. — Ambala, Haryana

- Analyzed 5G telecom network logs using SQL and Python to monitor tower traffic volume, identify call drop patterns, evaluate network performance, and support operational decision-making.
- Investigated traffic spikes, tower outage scenarios, and network congestion using telecom KPIs to assist traffic redistribution and maintain uninterrupted mobile network services.
- Processed and analyzed large network datasets to detect abnormal traffic behavior, identify congestion trends, and evaluate tower utilization across different network conditions.
- Generated analytical reports summarizing network performance, traffic anomalies, and operational insights to support network optimization and capacity planning.

Software Development Engineer (SDE) Intern Feb 2026 – Mar 2026

Bluestock Fintech — Remote

- Completed technical assessment with a score of 100/100 and contributed to React, Node.js, and Firebase-based applications with REST API integration.
- Developed UI components using React and Tailwind CSS while supporting feature implementation in a Git-based development environment.

Projects

RAG-Based AI Teaching Assistant Jan 2026 – Feb 2026

Python, LangChain, FAISS, OpenAI API, NLP

- Built a Retrieval-Augmented Generation (RAG) assistant using LangChain and FAISS to query 500+ documents, enabling accurate context-aware responses across 50+ queries per session.
- Designed an embedding-based semantic search pipeline that improved answer relevance by 30%, reduced hallucinations by 25%, and decreased query latency by 35%.
- Integrated OpenAI APIs with vector search, document chunking, and prompt engineering to deliver scalable document-based question answering.

Real-Time Anomaly Detection System Apr 2026 – May 2026

Python, Scikit-learn, Streamlit, Plotly

- Built a real-time anomaly detection system using Isolation Forest on 1000+ streaming data points for continuous monitoring of unusual patterns.
- Engineered time-series features and developed a Streamlit dashboard for live visualization and anomaly explanation, improving detection accuracy by 20%.

Leadership

Executive Core Member Feb 2024 – Jul 2025

Frosh (under the aegis of Dean of Student Affairs)

Thapar Institute of Engineering and Technology, Patiala

- Led **79** mentors as part of **20-member core team** handling **20,000+ applicants**.
- Coordinated **30-day Fresher's Fest** with **3,200+ participants**.
- Headed Media Department for event coverage and coordination.

Executive Committee Member — Saturnalia Tech Fest Sep 2025 – Nov 2025

- Led media operations for a 400+ member team, increased inter-college participation through outreach, and executed events with **3000+ attendees**.

Certifications

The Ultimate Job Ready Data Science Course — codewithharry.com Nov 2025 – Mar 2026

Decode DSA with C++ — PW Skills Mar 2026 – Jul 2026

Skills

Languages: Python, SQL, JavaScript, C++

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Streamlit

Tools: Git, GitHub, Jupyter Notebook, React.js, Firebase, Vercel

Core: Data Analysis, Machine Learning, EDA, Data Structures & Algorithms, REST APIs, RAG